

EXECUTIVE SUMMARY

The aim of the Guidelines is to promote Trustworthy AI. Trustworthy AI has **three components**, which should be met throughout the system's entire life cycle: (1) it should be **lawful**, complying with all applicable laws and regulations (2) it should be **ethical**, ensuring adherence to ethical principles and values and (3) it should be **robust**, both from a technical and social perspective since, even with good intentions, AI systems can cause unintentional harm. Each component in itself is necessary but not sufficient for the achievement of Trustworthy AI. Ideally, all three components work in harmony and overlap in their operation. If, in practice, tensions arise between these components, society should endeavour to align them.

These Guidelines set out a **framework for achieving Trustworthy AI**. The framework does not explicitly deal with Trustworthy AI's first component (lawful AI).¹ Instead, it aims to offer guidance on the second and third components: fostering and securing ethical and robust AI. Addressed to all stakeholders, these Guidelines seek to go beyond a list of ethical principles, by providing guidance on how such principles can be operationalised in socio-technical systems. Guidance is provided in three layers of abstraction, from the most abstract in Chapter I to the most concrete in Chapter III, closing with examples of opportunities and critical concerns raised by AI systems.

- I. Based on an approach founded on fundamental rights, **Chapter I** identifies the **ethical principles** and their correlated values that must be respected in the development, deployment and use of AI systems.

Key guidance derived from Chapter I:

- ✓ Develop, deploy and use AI systems in a way that adheres to the ethical principles of: *respect for human autonomy, prevention of harm, fairness and explicability*. Acknowledge and address the potential tensions between these principles.
- ✓ Pay particular attention to situations involving more vulnerable groups such as children, persons with disabilities and others that have historically been disadvantaged or are at risk of exclusion, and to situations which are characterised by asymmetries of power or information, such as between employers and workers, or between businesses and consumers.²
- ✓ Acknowledge that, while bringing substantial benefits to individuals and society, AI systems also pose certain risks and may have a negative impact, including impacts which may be difficult to anticipate, identify or measure (e.g. on democracy, the rule of law and distributive justice, or on the human mind itself.) Adopt adequate measures to mitigate these risks when appropriate, and proportionately to the magnitude of the risk.

- II. Drawing upon Chapter I, **Chapter II** provides guidance on how Trustworthy AI can be realised, by listing **seven requirements** that AI systems should meet. Both technical and non-technical methods can be used for their implementation.

Key guidance derived from Chapter II:

- ✓ Ensure that the development, deployment and use of AI systems meets the seven key requirements for Trustworthy AI: (1) human agency and oversight, (2) technical robustness and safety, (3) privacy and data governance, (4) transparency, (5) diversity, non-discrimination and fairness, (6) environmental and societal well-being and (7) accountability.
- ✓ Consider technical and non-technical methods to ensure the implementation of those requirements.

¹ All normative statements in this document aim to reflect guidance towards achieving the second and third component of trustworthy AI (ethical and robust AI). These statements are hence not meant to provide legal advice or to offer guidance on compliance with applicable laws, though it is acknowledged that many of these statements are to some extent already reflected in existing laws. In this regard, see §21 and following.

² See articles 24 to 27 of the Charter of Fundamental Rights of the EU (EU Charter), dealing with the rights of the child and the elderly, the integration of persons with disabilities and workers' rights. See also article 38 dealing with consumer protection.